MIL-B-46066B(MR)
16 June 1983
SUPERSEDING
MIL-B-46066A(MR)
31 July 1969

MILITARY SPECIFICATION

BANDS, ROTATING, COPPER ALLOY, CENTRIFUGAL CAST (FOR CRITICAL APPLICATIONS)

This specification is approved for use by the Army Materials and Mechanics Research Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one composition of copper alloy rotating bands produced by the centrifugal casting process.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

Fed. Test Method STd. No. 151 - Metals; Test Methods

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Materials and Mechanics Research Center, ATTN: DRXMR-SMS, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC NO. A3209

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45:22
Check the source to verify that this is the current version before use.

AREA MECA

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 8 - Tension Testing of Metallic Materials
E 10 - Brinell Hardness of Metallic Materials
E 290 - Semi-Guided Bend Test for Ductility of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI Standard B 46.1 - Surface Texture Surface Roughness, Waviness and Lay

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Chemical composition. The chemical composition shall meet the requirements shown in table I. The contractor shall furnish a certificate of compliance that castings meet the chemical requirements of table I.

Table I. Chemical composition

Element	Percent
Copper[a/]	89.0-93.0
Iron, maximum	0.15
Lead, maximum	0.050
Total elements other than copper or zinc, maximum	0.225
Zinc (by difference)	Remainder

a/ Silver to be counted as copper.

3.1.1 Check analysis. The chemical composition shall be subject to check analysis by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

3.2 Mechanical properties.

3.2.1 Tensile. Test specimens representing the bands shall meet the requirements shown in table II. Bands may be annealed to meet the mechanical properties.

Table II. Tensile properties

Tensile strength ksi	38
Elongation in 2 inches, percent	35

3.2.2 Bend. Test specimens representing the bands shall show no evidence of cracking on the outside of the bent portion when bent through an angle of 180 degrees with a radius equal to the thickness of the blank.

3.2.3 Hardness. The Brinell hardness number of the bands shall not be greater than BHN 55 when tested in accordance with 4.5.4 using a 500 kilogram load and 10 millimeter diameter ball.

3.3 Porosity and nonmetallic inclusion. Porosity and nonmetallic inclusions shall not be greater than that shown in figure 1 when examined at a magnification of 100X.

3.4 Copper-copper oxide eutectic. The bands shall be free of copper-copper oxide eutectic when examined at a magnification of 1000X. Copper-copper oxide eutectic is illustrated in figure 2.

3.5 Dimensions and tolerances. Dimensions and tolerances shall be as specified in the contract or order or on the applicable drawings (see 6.2).

3.6 Finish. Annealed bands, unless bright annealed, shall be pickled, washed, and dried, and shall be free from scale, chips, oil, grease, and dirt. Surface finish shall meet the requirements of 125 RMS when examined with a profilometer or by visual comparison methods (see 4.4.1).

3.7 Identification marking. In addition to any marking specified in the contract or order (see 6.2), all bands shall be identified by heat number and casting number if more than one casting is poured from a heat.

3.8 Workmanship. The bands shall be free from blowholes, hard spots, shrinkage defects, cracks, and other defects detrimental for the intended use.

Source: https://assist.dla.mil - Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot.

4.2.1 Visual, dimensional, and packaging. A lot for visual, dimensional, and packaging examination shall consist of all bands subjected to inspection at one time.

4.2.2 Testing. A lot for testing purposes shall consist of all bands poured from one heat or melt. When more than one melt is combined in a ladle, the contents of the ladle shall be regarded as one heat.

4.3 Sampling.

4.3.1 Chemical analysis. One sample shall be taken from each heat.

4.3.2 Mechanical properties. Two bands shall be selected from each lot for tensile, hardness, and bend testing.

4.3.3 Microporosity, nonmetallic inclusions, grain size. One sample shall be taken from each lot for testing for microporosity, nonmetallic inclusions, and grain size. These samples may be taken from the same bands used for mechanical properties testing.

4.3.4 Dimensions. Five bands shall be selected at random from each lot for dimensional examination.

4.4 Examination.

4.4.1 Visual. All bands shall be examined to determine compliance with the finish (see 3.6) and workmanship (see 3.8) requirements. Examination for finish shall be in accordance with ANSI B 46.1.

4.4.2 Dimensions. The samples shall be measured to determine compliance with the dimension and tolerance requirements (see 3.5).

4.4.3 Packaging inspection. Prior to shipment, examination shall be made of the preparation for shipment to determine compliance with section 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

4.5 Tests.

4.5.1 Chemical analysis. Chemical analysis shall be conducted in accordance with method 111 or 112 of Fed. Test Method Std. No. 151. In case of dispute, the analysis by method 111 shall be the basis for acceptance or rejection (see 3.1).

4.5.2 Tensile. Tensile test specimens shall be cut circumferentially from the sample bands and cold straightened by pressure, without hammering. Test specimens shall be prepared and tested in accordance with ASTM method E 8.

4.5.3 Bend. Bend test specimens shall consist of a section of the full width and thickness of the band with the subtended arc not less than 1/3 or more than 1/2 of the circumference of the band. Bend tests shall be conducted in accordance with ASTM method E 290.

4.5.4 Hardness. Hardness tests shall be conducted in accordance with ASTM method E 10.

4.5.5 Microporosity and nonmetallic inclusions. The specimen for microexamination for microporosity and nonmetallic inclusions shall be cut from the circumference of the band and shall represent the full thickness of the band. The specimen shall be prepared for microexamination in the following manner:

a. Grind face to a depth of 0.050 inch.

b. Polish successivly with 240, 320, and 400 grit silicon carbide papers.

c. Polish successively with 00 and 000 emery papers.

d. Polish initially on wheel using 3 microhn diamond paste with kerosene on broadcloth for 3 to 5 minutes at approximately 250 RPM.

e. Polish on final wheel using a water emulsion of gamma aluminum oxide less than 0.01 microhn on gamal cloth for 2 minutes at approximately 1150 RPM.

The unetched test specimen shall be examined at a magnification of 100X for compliance with the requirements of 3.3.

4.5.6 Copper-copper oxide eutectic. The unetched test specimen as described in 4.5.5 shall be examined at a magnification of 1000X for compliance with the requirements of 3.4.

4.6 Rejection and retest. Unless otherwise specified in the contract or order (see 6.2), rejection and retest shall be conducted in accordance with the general section of Fed. Test Method Std. No. 151.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation and packaging. Unless otherwise specified (see 6.2), preservation and packing shall be in accordance with level C.

5.1.1 Level C. Cleaning, drying, preservation, and packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing. Unless otherwise specified (see 6.2), packing shall be in accordance with level C.

5.2.1 Level C. The material shall be packed in a manner to insure carrier acceptance and safe delivery at destination.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The centrifugally-cast, copper-alloy rotating bands covered by this specification are intended for use in critical shell applications.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Dimensions and dimensional tolerances (see 3.5).
(c) Identification marking (see 3.7).
(d) Responsibility for inspection if other than shown in 4.1.
(e) Other rejection and retest requirements when applicable (see 4.6).
(f) Preservation, packaging, packing and marking requirements when different than shown in section 5.

6.2.2 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirement List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by the specification is cited in the following paragraphs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

Paragraph no.	Data requirement title	Applicable DID no.
3.1	Certificate of compliance	DI-E-2121

(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of the data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the Contracting Officer.)

6.3 Metric units. When metric dimensions are required, units for inch, and pounds per square inch may be converted to the metric equivalent by multiplying them by the following conversion factors:

English	Multiply by	Equals	Metric SI units
inch	0.0254	=	Metre (m)
pounds/sq in	$6.893X10^3$	=	Pascal (Pa)

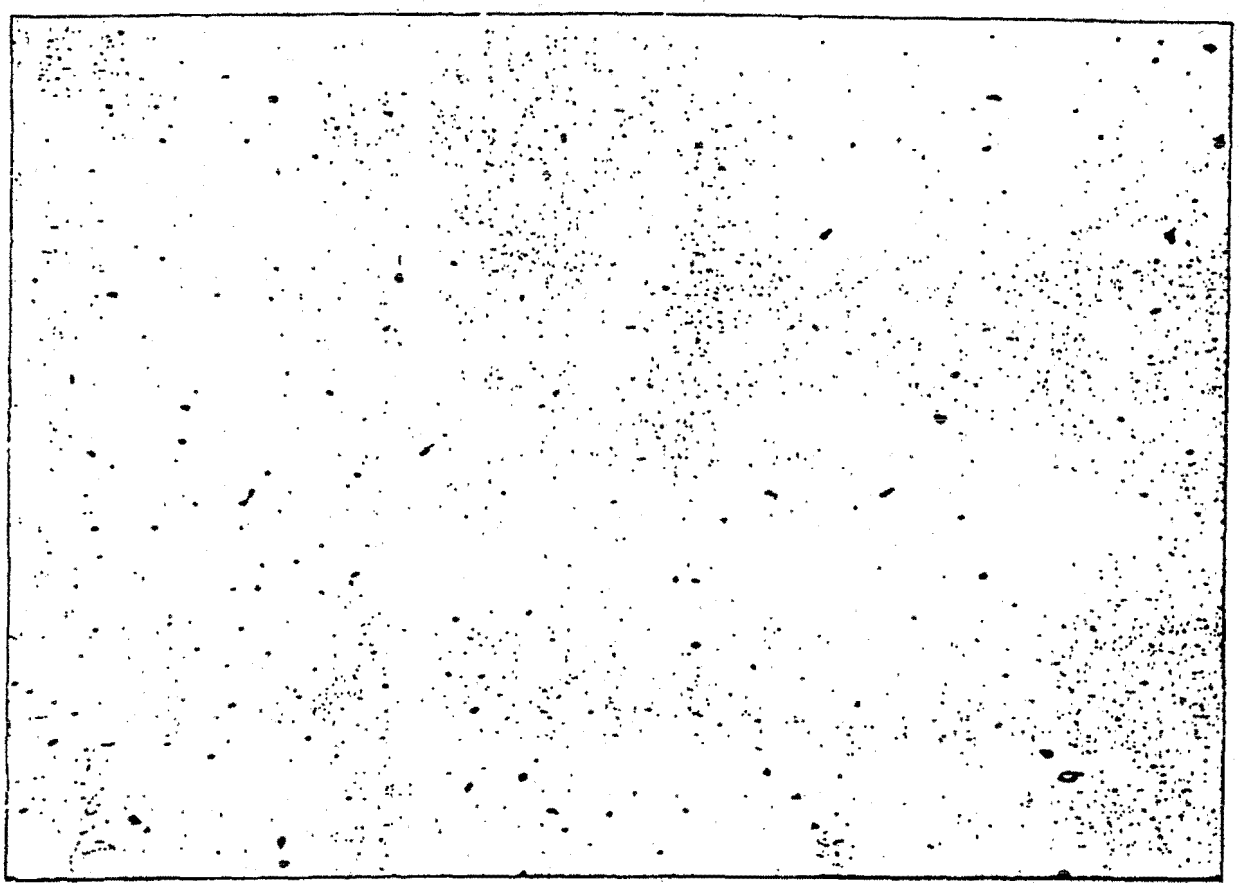
Custodian:
Army - MR

Review interest:
Army - AR

Preparing activity:
Army - MR
Project No. MECA-A089

(KBWP# ID-0055B/DISK 0139A. FOR AMMRC USE ONLY)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.

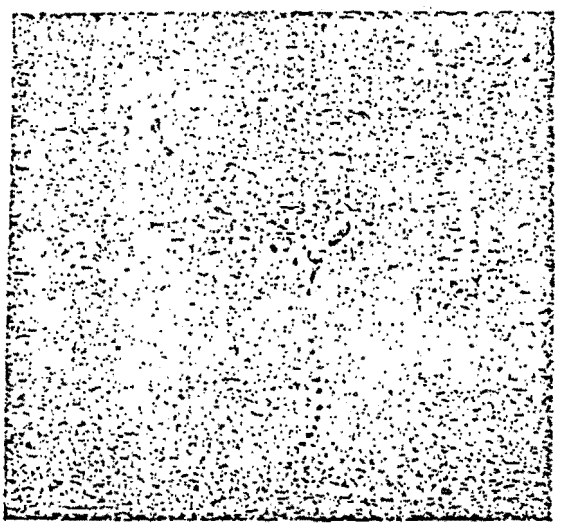


100X

FIGURE 1

MICROPOROSITY AND NONMETALLIC INCLUSIONS

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.



X1000

FIGURE 2

COPPER-COPPER OXIDE EUTECTIC

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:45Z
Check the source to verify that this is the current version before use.